

Abstract

Quantifying the Reliability of a Pneumonia-Diagnosing ResNet34 Model via GAX-based Interpretable Confidence Maps

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Deep learning models have demonstrated strong performance in medical image classification, yet their clinical reliability remains limited by the black-box problem and shortcut learning behavior, where models rely on medically irrelevant features instead of genuine pathological regions. This study proposes a quantitative reliability assessment framework using Generative Augmentative Explanations (GAX) and cheating scores (S_{Cheat}) to evaluate whether pneumonia classification models rely on clinically meaningful regions during prediction. Eight ResNet34 variants were trained on the RSNA Pneumonia Detection dataset for binary chest X-ray classification. GAX was utilized to generate interpretable confidence maps, while lung segmentation masks were applied to identify medically relevant pulmonary regions. A cheating score was then computed to quantify attribution intensity outside the lung regions, enabling measurement of shortcut-learning behavior. The study further analyzed the relationship between predictive performance and reliability using Pearson correlation analysis. Experimental results demonstrated that models with high classification accuracy may still exhibit strong reliance on medically irrelevant regions, highlighting that predictive performance alone is insufficient for assessing diagnostic reliability. The proposed framework contributes toward safer and more trustworthy deployment of deep learning systems in medical diagnosis through quantitative pre-deployment reliability assessment.